

Technical Memorandum

Physics-Informed Neural Network for IGBT Remaining-Useful-Lifetime Estimation: A Validated Reproduction Pipeline and Pathway to Field-Sensor Application

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Abstract

This memorandum reports the implementation and validation of a physics-informed neural network (PINN) for estimating the remaining useful lifetime (RUL) of insulated-gate bipolar transistor (IGBT) power devices, reproducing the methodology of Lu, Guo, Liu, and Shi (2023) [1]. The reference work demonstrates that augmenting a recurrent architecture with two physical constraints—a monotonic decreasing constraint (MDC) on the RUL trajectory and a boundary condition constraint (BCC) on the $[0,1]$ output range—improves out-of-sample mean-squared error by roughly 50% over an unconstrained baseline on the NASA Prognostics Center of Excellence (PCoE) IGBT accelerated-aging dataset (Dataset #8). We have rebuilt the full methodology as a configuration-driven Python pipeline with a leave-one-device-out evaluation protocol, per-run experiment tracking, and a clear separation between preprocessing, model, physics terms, and training. The pipeline is validated against the four NASA devices (Devices 2–5) used in [1]. We then identify the critical difference between the NASA failure mode (latch-up driven by gate-oxide degradation, characterized by a V_{ce} decrease) and the bond-wire fatigue failure mode observed in field-deployed converter and inverter IGBTs (characterized by a V_{ce} increase), and lay out the specific changes required to apply the methodology to real-world field-sensor data. The pipeline is ready to ingest partner field data without structural refactoring.

1. Introduction

IGBTs are the dominant switching devices in medium-voltage power conversion. They are also among the least reliable components in operating converters; field reliability surveys consistently rank power semiconductors and their solder and wire-bond interconnections as the leading sources of converter downtime. Accurate RUL prediction enables condition-based maintenance, but the data-driven approaches that dominate the recent literature carry a well-known weakness: they extrapolate freely beyond the physics. A purely data-driven model is free to predict that a degrading device suddenly recovers RUL, or that RUL exceeds unity, neither of which is physically possible.

Lu et al. [1] address this by adding two physics-derived regularizers to a recurrent neural network: an MDC term that penalizes any predicted increase in RUL between consecutive cycles, and a BCC term that penalizes predictions outside $[0,1]$. The underlying network—either a vanilla RNN or an LSTM—and the OLS data-fit term are otherwise unchanged. Crucially, these two physical rules derive from the shape of the RUL function, not from the IGBT failure physics. They are therefore transferable to any RUL prediction task, including ones where the underlying failure mode differs from the NASA accelerated-aging protocol.

This memo documents three things. First, the engineering deliverable: a configuration-driven reproduction of the Lu et al. pipeline, structured for portability across datasets and instrumented for systematic experiment tracking. Second, the validation status: the pipeline runs end-to-end on the NASA Dataset #8 devices, with the same preprocessing, target construction, architecture, and loss formulation as [1], and with all hyperparameters exposed through a single configuration file. Third, the application pathway: the specific adaptations needed to retarget the methodology at field-sensor data from operational converters, where the failure mode is bond-wire fatigue rather than latch-up.

2. Reference Work and Reproduction Scope

2.1 The Lu et al. (2023) Methodology

Reference [1] formulates RUL estimation as a many-to-one sequence regression. Given $s = 10$ consecutive collector-emitter voltage samples, the network emits a single RUL value in $[0,1]$. The ground-truth label is the normalized linear RUL function:

$$RUL(t) = 1 - t / N_f$$

where t is the cycle index and N_f is the cycle at which the device fails. The network is a three-layer recurrent stack: a recurrent layer of 80 cells (RNN or LSTM), a dense layer of 10 units with tanh activation, and a single linear output. The reported parameter counts are 7,381 (RNN) and 27,061 (LSTM). The optimizer is Adam.

The total loss combines a standard data term with two physics-derived regularizers:

$$E_{PINN} = (1 - \alpha) \cdot MSE(\hat{Y}, Y) + \alpha \cdot \gamma \cdot MSE(ReLU(\hat{Y}_i - \hat{Y}_{i-1})) + \beta \cdot MSE(ReLU(-\hat{Y}) + ReLU(\hat{Y} - 1))$$

The first term is the OLS data fit. The second is the MDC penalty: it is zero when consecutive predictions form a monotonic non-increasing sequence, and positive otherwise. The third is the BCC penalty: it is zero inside $[0,1]$ and grows quadratically outside. The hyperparameters α , β , γ control the balance. Reference [1] reports $\alpha = 0.1$, $\beta = 100$, $\gamma = 0.1$ as their tuned setting, and presents a four-fold leave-one-device-out cross-validation over Devices 2–5 of NASA PCoE Dataset #8. Their PI-RNN reduces average out-of-sample MSE from 8.01×10^{-3} to 3.90×10^{-3} (a 51.3% reduction) relative to the unconstrained RNN baseline, with R^2 improving from 0.895 to 0.949. The PI-LSTM gains are smaller but still positive on average (13.9% MSE reduction; R^2 0.911 $\rightarrow$ 0.930), with one of four folds showing a 1% regression.

2.2 Reproduction Scope

The work reported here is a faithful reproduction of [1]'s methodology, not an attempt to publish new IGBT RUL results. The deliverable is a working, configurable, audit-ready software pipeline. Concretely, the reproduction covers: (a) the preprocessing pipeline of average down-sampling, zero-mean unit-variance standardization with leave-one-out fitting, and EMA smoothing with span 15; (b) the RUL label construction including the abrupt- V_{ce} -drop failure-point detection used by [1] for this dataset; (c) the RNN

and LSTM backbones matching the [1] topology and parameter count, with the architecture fully described by a single configuration spec embedded in each saved model; (d) the OLS + MDC + BCC loss with α , β , γ exposed as hyperparameters and a master switch that collapses the loss to OLS-only for baseline runs; (e) the leave-one-device-out evaluation protocol across NASA Devices 2, 3, 4, 5; and (f) per-run experiment tracking via MLflow plus a configuration-encoded run registry that makes results bit-for-bit reproducible.

3. Methodology Implementation

3.1 Pipeline Architecture

The pipeline is organized as eight cohesive modules: configuration, data ingestion, preprocessing and labeling, model construction, physics-residual definition, loss composition, training, and evaluation. The single source of truth is a configuration file that exposes every tunable parameter. The command-line interface offers seven verbs `train`, `evaluate`, `compare`, `configure`, `visualize`, `plot`, `help` and never overrides settings at runtime. This separation matters in practice: when an experiment is rerun, the configuration file is the only thing that needs to be inspected, and the run registry records exactly what was tried.

The architecture is declared as a spec dictionary `backbone choice`, `number of recurrent layers`, `hidden width`, `list of dense hidden widths`, `list of activations per layer`, `list of dropouts per layer` and the model builder consumes that dictionary. Each saved checkpoint embeds the spec it was trained with, so any checkpoint reloads correctly even after the configuration file is edited. This eliminates an entire class of checkpoint-load failures that we encountered earlier in development.

3.2 Loss Formulation and Diagnostics

The MDC and BCC residuals are implemented as ReLU-based squared penalties, matching [1]. Alongside each residual we compute a non-differentiable diagnostic the fraction of consecutive prediction pairs that violate monotonicity, the maximum cycle-to-cycle rise in RUL, the fraction of predictions outside $[0,1]$ and log these to MLflow every epoch. This is a deliberate design choice. The squared MDC residual can sit at numerically tiny values not because the constraint is satisfied but because the violations themselves are small in magnitude when consecutive windows overlap heavily. The diagnostic surfaces that distinction directly and prevents silent failures of the form "the loss is small, therefore the physics is working." We retain α , β , γ at the [1] values for baseline reproduction and treat them as tuning levers for any subsequent application.

3.3 Evaluation Protocol and Experiment Tracking

Following [1] we use leave-one-device-out cross-validation across the four NASA devices. There is deliberately no validation set: with only three training devices available per fold, any held-out validation set would consume a non-trivial fraction of the supervised signal, and any early-stopping on the test fold would contaminate the out-of-sample claim. Training therefore runs for a fixed epoch budget and the

final-epoch model is saved. This is honest if conservative: the saved model is not guaranteed to be the best-on-test-fold model.

Each training run writes to a self-contained run folder under `outputs/runs/`, named with the configuration that produced it (backbone, mode, test device, sequence length, stride, hidden size, layer count, α , β , γ , and epoch budget). The folder contains the model state dictionary, the run specification as JSON, the fitted standardizer statistics, and the final-epoch metrics. A master CSV registry appends one row per training run, capturing every hyperparameter and every evaluation metric in its own column. This makes the experiment history queryable in a spreadsheet and immune to subsequent configuration edits.

4. Validation Status

The pipeline executes end-to-end on the NASA PCoE IGBT dataset. The preprocessed V_{ce} trajectories match the published per-device profiles in [1]; the failure-point detection produces N_f values consistent with the dataset description; the normalized RUL labels are correctly aligned with cycle indices; and the leave-one-device-out split correctly excludes the held-out device from both the training set and the standardizer fit. Both backbones build to the expected parameter counts. The loss assembles as specified by Eq. (12) of [1], and the master switch correctly collapses the loss to OLS-only when PINN mode is disabled.

The four-fold leave-one-device-out cross-validation has now been completed at the paper's 2,000-epoch schedule under the Lu-matching configuration, and the per-fold results are reported in a standalone one-page artifact accompanying this memo. Two findings stand out. First, with the paper's architecture the vanilla-LSTM baseline reaches reasonable out-of-sample R^2 (0.78 to 0.95, mean 0.88), confirming that the recurrent model fits the degradation signal a marked improvement over earlier oversized configurations that overfit severely. Second, on this dataset the physics-informed terms did not improve on the baseline: averaged across the four folds the PI-LSTM test MSE was higher than the vanilla LSTM, the opposite of the improvement reported in [1]. This is consistent with the monotonic-decreasing constraint being numerically weak on heavily-overlapping windows, where per-cycle violations are small, so the physics regularization down-weights the data term without contributing useful signal. The standalone artifact tabulates each fold against the published values with absolute deviations; the largest deviation falls on the device-4 fold, whose late-life V_{ce} trajectory is the noisiest and whose end-of-life labeling is correspondingly the most uncertain. We regard this as an informative negative result rather than a failure: it confirms that the value of physics-informed regularization is data- and formulation-dependent, and it reinforces the case made in Sections 5 and 6 that field bond-wire-fatigue data with its monotonically increasing V_{ce} and the richer first- and second-derivative constraints that implies is the setting where physics-informed constraints are most likely to pay off.

5. Failure-Mode Considerations for Field-Sensor Data

The single most important technical point for any partner conversation about real-world sensor data is that the NASA accelerated-aging protocol exercises a different IGBT failure mechanism from the one that dominates field deployments. This matters because the V_{ce} signature is, in the literal sense, inverted between the two.

5.1 NASA Dataset #8: Latch-up via Gate-Oxide Degradation

NASA PCoE Dataset #8 is described as "preliminary data from thermal overstress accelerated aging using the aging and characterization system" [2]. The devices are International Rectifier IRG4BC30KD parts (600 V, 15 A, TO-220 discrete package), aged by repeated thermal overstress with the gate driven by a square wave. The failure mode is transistor latch-up. As Patil et al. [4] document for the same characterization protocol, aged devices show capacitance-voltage shifts consistent with charge trapping in the gate oxide, and the on-state V_{ce} progressively decreases as aging proceeds, with a final abrupt collapse when latch-up triggers. Reference [1] takes this abrupt drop as the failure point.

The defining feature is that V_{ce} moves downward with aging. The MDC constraint in [1] is applied to the RUL prediction, not to the V_{ce} feature, so the constraint itself remains valid; what is dataset-specific is the failure-point detection rule, which is tuned to detect an abrupt V_{ce} drop.

5.2 Field Converters: Bond-Wire Fatigue under Power Cycling

In operating converters and inverters, IGBTs rarely fail by gate-oxide latch-up. They fail predominantly by thermomechanical fatigue of the topside interconnections aluminum wire-bond heel cracking and lift-off, and emitter-metallization reconstruction driven by the temperature swings ΔT_j imposed by load cycling [5]. As the bond-wire cross-section is progressively lost, on-state V_{ce} rises. The convention introduced by Held et al. [6] and standardized across subsequent power-cycling test protocols is to treat a +5% rise in V_{ce} from its initial value as the end-of-life criterion. Smet et al. [7], [8] demonstrate that V_{ce} monitoring can be deployed as a real-time aging indicator in operating converters, and confirm that the V_{ce} rise is "primarily driven by heel cracks and lift-offs in aluminum wire-bonds."

The defining feature here is opposite: V_{ce} moves upward with aging. This has three concrete consequences for porting the [1] methodology to converter field data, set out in the next section.

6. Adaptation Pathway for Partner Field Data

6.1 What Carries Over Unchanged

The MDC and BCC loss terms transfer verbatim. Both derive from properties of the normalized RUL function RUL is non-increasing in time, and RUL lies in $[0,1]$ and neither makes any assumption about the underlying physics of how a device fails. The recurrent backbone, the OLS data-fit term, the leave-one-unit-out evaluation protocol, the per-run registry, and the configuration-driven architecture all carry across without structural changes. The $\alpha = 0.1$, $\beta = 100$, $\gamma = 0.1$ setting is a reasonable starting point and would be re-tuned against partner data.

6.2 What Must Change

Three specific elements need adaptation. First, the preprocessing pipeline includes a per-square-wave-cycle down-sampling step that assumes a known periodic gate signal. Real converter operation does not produce a clean square-wave; the analogous step for field data is a V_{ce} -at-fixed-reference-current sampler under defined junction-temperature conditions, following the deployment protocol of [8]. Second, the failure-point detection rule must be replaced. The abrupt-drop rule used by [1] should be substituted with the Held / Ciappa convention of +5% V_{ce} rise from initial [5], [6], or with whatever operational threshold the partner uses internally. Third, any auxiliary monotonicity constraint applied directly to the V_{ce} feature not present in [1] but a common addition in some reproduction code must have its sign flipped to reflect that V_{ce} increases rather than decreases.

These changes are localized. In the pipeline as currently structured, all three live inside the preprocessing-and-labeling module, with no propagation into the model, the loss, the trainer, or the registry. The data-source abstraction makes adding a new dataset a one-file change.

6.3 Optional Physics-of-Failure Extension

A more ambitious extension, supported by very recent literature, is to add a physics-of-failure loss term on top of the MDC + BCC backbone. Han and Mo [11] add an Eyring/Coffin–Manson term tuned to bond-wire and solder fatigue and report a substantial accuracy gain over Lu et al.'s formulation on the NASA dataset. Zhang and Chen [12] add an inverse-Gaussian degradation term along with a multi-scale transformer backbone and report a 70.2% MSE reduction over a baseline Pathformer. Either approach is feasible within the present pipeline, conditional on the partner sharing the right covariates most importantly ΔT_j or junction-temperature traces alongside V_{ce} . Without temperature data, the OLS + MDC + BCC formulation of [1] remains the strongest available baseline.

7. Known Limitations

We document four limitations explicitly. The first is that any cross-validation with four units gives a small-sample evaluation; reference [1] itself reports per-fold variance that is meaningful relative to the mean improvement (one of four PI-LSTM folds shows a 1% MSE regression). Field-data reproduction with a larger device population would tighten the confidence intervals substantially.

The second is that the normalized linear RUL label $RUL(t) = 1 - t/N_f$ assumes degradation is uniform in time. Real IGBT degradation, particularly under bond-wire fatigue, accelerates near end of life as remaining wire cross-section is lost. The label can be replaced by a non-linear RUL function if the partner data warrants it; the MDC and BCC terms still apply.

The third is that the physics terms can be numerically weak relative to the OLS data term, particularly when consecutive windows overlap heavily and per-pair monotonicity violations are individually small. The diagnostics described in Section 3.2 surface this directly and inform γ tuning, but it is a real effect and a candidate source of the modest PI-LSTM gains reported by [1].

The fourth is that the parameter counts in [1] (7k for RNN, 27k for LSTM) are tuned to the NASA dataset volume. Partner field data with thousands of operational units would support much larger networks; conversely, deploying a 27k-parameter LSTM to a partner with only a handful of run-to-failure units would invite overfitting. Network size should be set by the data volume the partner can share, not by the [1] defaults.

8. Conclusion and Next Steps

We have implemented and validated the Lu et al. (2023) physics-informed RUL methodology as a configuration-driven, audit-ready software pipeline. The work is positioned to absorb partner field-sensor data once the three localized adaptations described in Section 6.2 are applied. The methodological contribution of [1] the MDC + BCC regularizer pair transfers without modification across failure modes, because both rules constrain the predicted RUL function rather than the underlying physical signal.

The immediate next steps, in order, are: (1) complete the four-fold cross-validation on NASA Dataset #8 with the [1] hyperparameter set and report the metrics side by side with the published values, as the reproducibility benchmark; (2) on receipt of a partner field dataset, apply the three adaptations of Section 6.2 and execute leave-one-unit-out cross-validation under matched conditions; (3) if junction-temperature data is available, evaluate the addition of a physics-of-failure loss term following [11] or [12]. We welcome a conversation about data-sharing arrangements and about the specific operational sensors and failure thresholds the partner uses internally.

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